

Sirui She

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Education

University of California, San Diego

B.S. Mathematics–Computer Science

La Jolla, CA

Expected June 2027

Relevant Coursework: Object-Oriented Programming C++, Java, Linear Algebra, Probability, Discrete Mathematics, Graph Theory

Experience

STEM Math Tutor

Pasadena City College – Math Success Center

Pasadena, CA

Sep 2024 – Jun 2025

- Provided bilingual (English–Mandarin) tutoring in Calculus, Linear Algebra, and Differential Equations to 100+ students.
- Explained abstract mathematical concepts through structured problem-solving strategies and clear verbal reasoning.
- Supported students from diverse academic backgrounds, strengthening communication and teaching skills.

Projects

SlangSnap

Dec 2025 – Present

- Developed a cross-platform mobile application in **Flutter/Dart**; leveraged **Cursor** for rapid UI prototyping and **Xcode** for iOS builds, debugging, and simulator testing.
- Implemented a **modular architecture** separating **UI components**, **data models**, and **service-layer logic** to support scalable feature development.
- Integrated **curated datasets** and **external APIs**, applying **rule-based filtering** and **ranking heuristics** to improve data quality.
- Planning **ML-based recommendation methods** using **TF-IDF**, **cosine similarity**, and **collaborative filtering**.

CNN Image Classifier

Aug 2025

- Designed and trained a **Convolutional Neural Network (CNN)** using **TensorFlow** and **Keras**, applying **convolution**, **pooling**, and **fully connected layers**.
- Optimized model performance through **hyperparameter tuning** (learning rate, batch size) and **regularization**.
- Implemented **data preprocessing** and **augmentation pipelines** using **NumPy** and **OpenCV**.

Skills

Programming: Python, C++, Java, Dart

Machine Learning: TensorFlow, Keras, NumPy, OpenCV

Software Tools: Git/GitHub, CMake, Xcode, Cursor

Mathematics: Linear algebra, probability, optimization, discrete mathematics, graph algorithms

Leadership: Coordinated bilingual volunteer programs connecting students with local schools; experience in team organization and cross-cultural collaboration